

Exercise 2.8.4. Give a different proof⁽³³⁾ of the mean ergodic theorem (Theorem 2.21) as follows. For a measure-preserving system $(X, \mathcal{B}, \mu, T)$ and function $f \in L^2_\mu$, show that the function $n \mapsto \langle U_T^n f, f \rangle$ is positive-definite (see Sect. C.3). Apply the Herglotz–Bochner theorem (Theorem C.9) to translate the problem into one concerned with functions on $\mathbb{S}^1$, and there use the fact that $\frac{1}{N} \sum_{n=1}^N \rho^n$ converges for $\rho \in \mathbb{S}^1$ (to zero, unless $\rho = 1$).

2.9 Induced Transformations

Poincaré recurrence gives rise to an important inducing construction introduced by Kakutani [172]. Throughout this section, $(X, \mathcal{B}, \mu, T)$ denotes an invertible measure-preserving system⁽³⁴⁾.

Let $(X, \mathcal{B}, \mu, T)$ be an invertible measure-preserving system, and let A be a measurable set with $\mu(A) > 0$. By Poincaré recurrence, the first return time to A , defined by

$$r_A(x) = \inf_{n \geq 1} \{n \mid T^n(x) \in A\} \quad (2.36)$$

exists (that is, is finite) almost everywhere.

Definition 2.42. The map $T_A : A \rightarrow A$ defined (almost everywhere) by

$$T_A(x) = T^{r_A(x)}(x)$$

is called the transformation *induced* by T on the set A .

Notice that both $r_A : X \rightarrow \mathbb{N}$ and $T_A : A \rightarrow A$ are measurable by the following argument. For $n \geq 1$, write $A_n = \{x \in A \mid r_A(x) = n\}$. Then the sets

$$\begin{aligned} A_1 &= A \cap T^{-1}A, \\ A_2 &= A \cap T^{-2}A \setminus A_1, \\ &\vdots \\ A_n &= A \cap T^{-n}A \setminus \bigcup_{i < n} A_i \end{aligned}$$

are all measurable, as is

$$T^n A_n = A \cap T^n A \setminus (TA \cup T^2 A \cup \dots \cup T^{n-1} A),$$

since T is invertible by assumption.

Lemma 2.43. *The induced transformation T_A is a measure-preserving transformation on the space $(A, \mathcal{B}|_A, \mu_A = \frac{1}{\mu(A)}\mu|_A, T_A)$. If T is ergodic with respect to μ then T_A is ergodic with respect to μ_A .*

The notation means that the σ -algebra consists of $\mathcal{B}|_A = \{B \cap A \mid B \in \mathcal{B}\}$ and the measure is defined for $B \in \mathcal{B}|_A$ by $\mu_A(B) = \frac{1}{\mu(A)}\mu(B)$. The effect of T_A is seen in the *Kakutani skyscraper* Fig. 2.2. The original transformation T sends any point with a floor above it to the point immediately above on the next floor, and any point on a top floor is moved somewhere to the base floor A . The induced transformation T_A is the map defined almost everywhere on the bottom floor by sending each point to the point obtained by going through all the floors above it and returning to A .

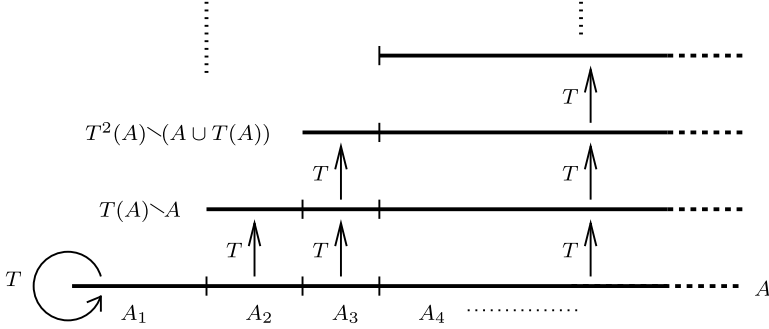


Fig. 2.2 The induced transformation T_A

PROOF OF LEMMA 2.43. If $B \subseteq A$ is measurable, then $B = \bigsqcup_{n \geq 1} B \cap A_n$ is a disjoint union so

$$\mu_A(B) = \frac{1}{\mu(A)} \sum_{n \geq 1} \mu(B \cap A_n). \quad (2.37)$$

Now

$$T_A(B) = \bigsqcup_{n \geq 1} T_A(B \cap A_n) = \bigsqcup_{n \geq 1} T^n(B \cap A_n),$$

so

$$\begin{aligned} \mu_A(T_A(B)) &= \frac{1}{\mu(A)} \sum_{n \geq 1} \mu(T^n(B \cap A_n)) \\ &= \frac{1}{\mu(A)} \sum_{n \geq 1} \mu(B \cap A_n) \quad (\text{since } T \text{ preserves } \mu) \\ &= \mu(B) \end{aligned}$$

by (2.37).

If T_A is not ergodic, then there is a T_A -invariant measurable set $B \subseteq A$ with $0 < \mu(B) < \mu(A)$; it follows that $\bigcup_{n \geq 1} \bigcup_{j=0}^{n-1} T^j(B \cap A_n)$ is a non-trivial T -invariant set, showing that T is not ergodic. $\square$

Poincaré recurrence (Theorem 2.11) says that for any measure-preserving system $(X, \mathcal{B}, \mu, T)$ and set A of positive measure, almost every point on the ground floor of the associated Kakutani skyscraper returns to the ground floor at some point. Ergodicity strengthens this statement to say that almost every point of the entire space X lies on some floor of the skyscraper. This enables a quantitative version of Poincaré recurrence to be found, a result due to Kac [168].

Theorem 2.44 (Kac). *Let $(X, \mathcal{B}, \mu, T)$ be an ergodic measure-preserving system and let $A \in \mathcal{B}$ have $\mu(A) > 0$. Then the expected return time to A is $\frac{1}{\mu(A)}$; equivalently*

$$\int_A r_A \, d\mu = 1.$$

PROOF⁽³⁵⁾. Referring to Fig. 2.2, each column

$$A_n \sqcup T(A_n) \sqcup \cdots \sqcup T^{n-1}(A_n)$$

comprises n disjoint sets each of measure $\mu(A_n)$, and the entire skyscraper contains almost all of X by ergodicity and Proposition 2.14(3) applied to the transformation T^{-1} . It follows that

$$1 = \mu(X) = \sum_{n \geq 1} n\mu(A_n) = \int_A r_A \, d\mu$$

by the monotone convergence theorem (Theorem A.16), since r_A is the increasing limit of the functions $\sum_{k=1}^n k\chi_{A_k}$ as $n \rightarrow \infty$. $\square$

Kakutani skyscrapers are a powerful tool in ergodic theory. A simple application is to prove the Kakutani–Rokhlin lemma (Lemma 2.45) proved by Kakutani [172] and Rokhlin [315].

Lemma 2.45 (Kakutani–Rokhlin). *Let $(X, \mathcal{B}, \mu, T)$ be an invertible ergodic measure-preserving system and assume that μ is non-atomic (that is, $\mu(\{x\}) = 0$ for all $x \in X$). Then for any $n \geq 1$ and $\varepsilon > 0$ there is a set $B \in \mathcal{B}$ with the property that*

$$B, T(B), \dots, T^{n-1}(B)$$

are disjoint sets, and

$$\mu(B \sqcup T(B) \sqcup \cdots \sqcup T^{n-1}(B)) > 1 - \varepsilon.$$

As the proof will show, the lemma uses only division (constructing a quotient and remainder) and the Kakutani skyscraper.

PROOF OF LEMMA 2.45. Let A be a measurable set with $0 < \mu(A) < \varepsilon/n$ (such a set exists by the assumption that μ is non-atomic) and form the Kakutani skyscraper over A . Then X decomposes into a union of disjoint columns of the form

$$A_k \sqcup T(A_k) \sqcup \cdots \sqcup T^{k-1}(A_k)$$

for $k \geq 1$, as in Fig. 2.2. Now let

$$B = \bigsqcup_{k \geq n} \bigsqcup_{j=0}^{\lfloor k/n \rfloor - 1} T^{jn}(A_k),$$

the set obtained by grouping together that part of the ground floor made up of the sets A_k with $k \geq n$ together with every n th floor above that part of the ground floor (stopping before the top of the skyscraper). By construction the sets $B, T(B), \dots, T^{n-1}(B)$ are disjoint, and together they cover all of X apart from a set comprising no more than n of the floors in each of the towers, which therefore has measure no more than $n \sum_{k=1}^{\infty} \mu(A_k) \leq n\mu(A) < \varepsilon$. $\square$

One often refers to the structure given by Lemma 2.45 as a *Rokhlin tower* of height n with base B and residual set of size ε .

Exercises for Sect. 2.9

Exercise 2.9.1. Show that the inducing construction can be reversed in the following sense. Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system, and let $r : X \rightarrow \mathbb{N}_0$ be a map in L^1_μ . The *suspension* defined by r is the system $(X^{(r)}, \mathcal{B}^{(r)}, \mu^{(r)}, T^{(r)})$, where:

- $X^{(r)} = \{(x, n) \mid 0 \leq n < r(x)\}$;
- $\mathcal{B}^{(r)}$ is the product σ -algebra of $\mathcal{B}$ and the Borel σ -algebra on $\mathbb{N}$ (which comprises all subsets);
- $\mu^{(r)}$ is defined by $\mu^{(r)}(A \times N) = \frac{1}{\int r d\mu} \mu(A) \times |N|$ for $A \in \mathcal{B}$ and $N \subseteq \mathbb{N}$; and
- $T^{(r)}(x, n) = \begin{cases} (x, n+1) & \text{if } n+1 < r(x); \\ (T(x), 0) & \text{if } n+1 = r(x). \end{cases}$

(a) Verify that this defines a finite measure-preserving system.

(b) Show that the induced map on the set $A = \{(x, 0) \mid x \in X\}$ is isomorphic to the original system $(X, \mathcal{B}, \mu, T)$.

Exercise 2.9.2. ⁽³⁶⁾ The hypothesis of ergodicity in Lemma 2.45 can be weakened as follows. An invertible measure-preserving system $(X, \mathcal{B}, \mu, T)$ is called *aperiodic* if $\mu(\{x \in X \mid T^k(x) = x\}) = 0$ for all $k \in \mathbb{Z} \setminus \{0\}$.

(a) Show that an ergodic transformation on a non-atomic space is aperiodic.

(b) Find an example of an aperiodic transformation on a non-atomic space that is not ergodic.

(c) Prove Lemma 2.45 for an invertible aperiodic transformation on a non-atomic space.

Exercise 2.9.3. ⁽³⁷⁾ Show that the Kakutani–Rokhlin lemma (Lemma 2.45) does not hold for arbitrary sequences of iterates of the map T . Specifically, show that for an ergodic measure-preserving system $(X, \mathcal{B}, \mu, T)$, sequence $a_1, \dots, a_n$ of distinct integers, and $\varepsilon > 0$ it is not always possible to find a measurable set A with the properties that $T^{a_1}(A), \dots, T^{a_n}(A)$ are disjoint and $\mu(\bigcup_{i=1}^n T^{a_i}(A)) > \varepsilon$.

Exercise 2.9.4. Use Exercise 2.9.2 above to prove the following result of Steele [351]. Let $(X, \mathcal{B}, \mu, T)$ be an invertible aperiodic measure-preserving system on a non-atomic space. Then, for any $\varepsilon > 0$, there is a set $A \in \mathcal{B}$ with $\mu(A) < \varepsilon$ with the property that for any finite set $F \subseteq X$, there is some $j = j(F)$ with $F \subseteq T^{-j}(A)$.

Notes to Chap. 2

⁽¹²⁾(Page 16) A measurable isomorphism is also sometimes called a *conjugacy*; conjugacy is also used to describe an isomorphism between the measure algebras that implies isomorphism on sufficiently well-behaved probability spaces. This is discussed in Walters [374, Sect. 2.2] and Royden [320].

⁽¹³⁾(Page 17) The shift maps constructed here are measure-preserving transformations, but they are also homeomorphisms of a compact metric space in a natural way. The study of the dynamics of closed shift-invariant subsets of these systems comprises *symbolic dynamics* and is a rich theory in itself. A gentle introduction may be found in the book of Lind and Marcus [230] or Kitchens [197]; further reading in the collection edited by Berthé, Ferenczi, Mauduit and Siegel [93].

⁽¹⁴⁾(Page 21) Poincaré’s formulation in [288, Th. I, p. 69] is as follows:

“Supposons que le point P reste à distance finie, et que le volume

$$\int dx_1 dx_2 dx_3$$

soit un invariant intégral; si l’on considère une région r_0 quelconque, quelque petite que soit cette région, il y aura des trajectoires qui la traverseront une infinité de fois. [...] En effet le point P restant à distance finie, ne sortira jamais d’une région limitée R .”

The modern abstract measure-theoretic statement in Theorem 2.11 appears in a paper of Carathéodory [49].

⁽¹⁵⁾(Page 23) The notion of ergodicity predates the ergodic theorems of the 1930s, in various guises. These include the seminal work of Borel [40], described by Doob as being

“characterized by convenient neglect of error terms in asymptotics, incorrect reasoning, and correct results,”

as well as that of Knopp [205]; a striking remark of Novikoff and Barone [273] is that a result implicit in the work of van Vleck [370] on non-measurable subsets of $[0, 1]$ is that any measurable subset of $[0, 1]$ invariant under the map $x \mapsto 2x \pmod{1}$ has measure zero or one, a prototypical ergodic statement. The general formulation was given by Birkhoff and Smith [35].

⁽¹⁶⁾(Page 28) These operators are usually called Koopman operators; Koopman [208] used the then-recent development of functional analysis and Hilbert space by von Neumann [266] and Stone [354] to use these operators in the setting of flows arising in classical Hamiltonian mechanics.

⁽¹⁷⁾(Page 30) Even though this is not necessary here, we assume for simplicity that Hilbert spaces are separable, and as a result that they have countable orthonormal bases. As discussed in Sect. A.6, we only need the separable case.

⁽¹⁸⁾(Page 32) For a recent account of the history of the relationship between the two results and the account of how they came to be published as and when they did, see Zund [395]. The issue has also been discussed by Ulam [365] and others. The note [25] by Bergelson discusses both the history and how the two results relate to more recent developments.

⁽¹⁹⁾(Page 35) This result is simply one of many extensions and generalizations of the mean ergodic theorem (Theorem 2.21) to other complete function spaces. It is a special instance of the mean ergodic theorem for Banach spaces, due to Kakutani and Yosida [171, 391, 392].

⁽²⁰⁾(Page 38) The maximal ergodic theorem is due to Wiener [382] and was also proved by Yosida and Kakutani [392].

⁽²¹⁾(Page 40) Covering lemmas of this sort were introduced by Vitali [369], and later became important tools in the proof of the Hardy–Littlewood maximal inequality, and thence of the Lebesgue density and differentiation theorems (Theorems A.24 and A.25).

⁽²²⁾(Page 44) Birkhoff based his proof on a weaker maximal inequality concerning the set of points on which $\limsup_{n \rightarrow \infty} A_n^f \geq \alpha$, and initially formulated his result for indicator functions in the setting of a closed analytic manifold with a finite invariant measure. Khinchin [189] showed that Birkhoff’s result applies to integrable functions on abstract finite measure spaces, but made clear that the idea of the proof is precisely that used by Birkhoff. A natural question concerning Theorem 2.30, or indeed any convergence result, is whether anything can be said about the rate of convergence. An important special case is the law of the iterated logarithm due to Hartman and Wintner [141]: if $\|f\|_2 = 1$, $\int f \, d\mu = 0$ and the functions $f, U_T f, U_T^2 f, \dots$ are all independent, then

$$\limsup_{n \rightarrow \infty} A_n^f / \sqrt{(2 \log \log n)/n} = 1$$

almost everywhere (and $\liminf = -1$ by symmetry). It follows that

$$A_n^f = O\left(\left(\frac{1}{n} \log \log n\right)^{1/2}\right)$$

almost everywhere. However, the hypothesis of independence is essential: Krengel [210] showed that for any ergodic Lebesgue measure-preserving transformation T of $[0, 1]$ and sequence $(a_n) \rightarrow 0$ as $n \rightarrow \infty$, there is a continuous function $f : [0, 1] \rightarrow \mathbb{R}$ for which

$$\limsup_{n \rightarrow \infty} \frac{1}{a_n} \left| A_n^f - \int f \, d\mu \right| = \infty$$

almost everywhere, and

$$\limsup_{n \rightarrow \infty} \frac{1}{a_n} \left\| A_n^f - \int f \, d\mu \right\|_p = \infty$$

for $1 \leq p \leq \infty$. An extensive treatment of ergodic theorems may be found in the monograph of Krengel [211].

Despite the absence of any general rate bounds in the ergodic theorem, the constructive approach to mathematics has produced rate results in a different sense, which may lead to effective versions of results like the multiple recurrence theorem. Bishop's work [36] included a form of ergodic theorem, and Spitters [348] found constructive characterizations of the ergodic theorem. As an application of 'proof mining', Avigad, Gerhardy and Towsner [12] gave bounds on the rate of convergence that can be explicitly computed in terms of the initial data (T and f) under a weak hypotheses, while earlier work of Simic and Avigad [13, 346] showed that, in general, it is impossible to compute such a bound. An overview of this area and its potential may be found in the survey [11] by Avigad.

(23)(Page 44) Despite the impressive result in Example 2.31, the numbers known to be normal to every base have been constructed to meet the definition of normality (with the remarkable exception of Chaitin's constant [53]). Champernowne [54] showed that the specific number $0.123456789101112131415\dots$ is normal in base 10, and Sierpiński [345] constructed a number normal to every base. Sierpiński's construction was reformulated to be recursive by Becher and Figueira [20], giving a computable number normal to every base. The irrational numbers arising naturally in other fields, like π , e , $\zeta(3)$, $\sqrt{2}$, and so on, are not known to be normal to any base.

(24)(Page 44) There are many proofs of the pointwise ergodic theorem; in addition to that of Birkhoff [33] there is a more elementary (though intricate) argument due to Katznelson and Weiss [186], motivated by a paper of Kamae [177]. A different proof is given by Jones [167].

(25)(Page 50) This conjectured result—the “Rokhlin problem”—has been shown in important special cases by Host [158], Kalikow [176], King [193], Ryzhikov [328], del Junco and Yassawi [68, 390] and others, but the general case is open.

(26)(Page 50) The definition used by Koopman and von Neumann is the spectral one that will be given in Theorem 2.36(5), and was called by them the absence of “angle variables”; they also considered flows (measure-preserving actions of $\mathbb{R}$ rather than actions of $\mathbb{Z}$ or $\mathbb{N}$). In physical terms, they characterized lack of ergodicity as barriers that are never passed, and the presence of an angle variable as a clock that never changes, under the dynamics.

(27)(Page 50) Examples of such systems were constructed using Gaussian processes by Maruyama [255]; Kakutani [174] gave a direct combinatorial construction of an example (this example is described in detail in the book of Petersen [282, Sect. 4.5]). Other examples were found by Chacon [51, 52] and Katok and Stepin [185]. Indeed, there is a reasonable way of viewing the collection of all measure-preserving transformations of a fixed space in which a typical transformation is weak-mixing but not mixing (see papers of Rokhlin [315] and Halmos [135] or Halmos' book [138, pp. 77–80]).

(28)(Page 52) This more subtle version of Exercise 2.7.1 appears in a paper of Halmos [136], and is attributed to Ambrose, Halmos and Kakutani in Petersen's book [282].

(29)(Page 54) This is shown in the notes of Halmos [138]. Ergodicity also makes sense for transformations preserving an infinite measure; in that setting Kakutani and Parry [175] used random walk examples of Gillis [115] to show that for any $k \geq 1$ there is an infinite measure-preserving transformation T with $T^{(k)}$ ergodic and $T^{(k+1)}$ not ergodic.

(30)(Page 54) This is also known as exponential or effective rate of mixing or decay of correlations; see Baladi [15] for an overview of dynamical settings where it is known.

(31)(Page 58) A more constructive proof of the difficult step in Theorem 2.36 (which may be taken to be (5) $\implies$ (1)) exploiting properties of almost-periodic functions on compact groups, and giving more insight into the structure of ergodic measure-preserving transformations that are not weak-mixing, may be found in Petersen [282, Sect. 4.1].

(32)(Page 59) This is an example of a Hilbert–Schmidt operator [331]; a convenient source for this material is the book of Rudin [321] or Appendix B.

(33)(Page 61) This way of viewing ergodic theorems lies at the start of a sophisticated investigation of ergodic theorems along arithmetic sets of integers by Bourgain [41]. This

exercise already points at a relationship between ergodic theorems and equidistribution on the circle.

⁽³⁴⁾(Page 61) Notice that the assumption that $(X, \mathcal{B}, \mu, T)$ is invertible also implies that T is *forward measurable*, that is $T(A) \in \mathcal{B}$ for any $A \in \mathcal{B}$. Heinemann and Schmitt [146] prove the Rokhlin lemma for an aperiodic measure-preserving transformation on a Borel probability space using Exercise 5.3.2 and Poincaré recurrence instead of a Kakutani tower (aperiodic is defined in Exercise 2.9.2; for Borel probability space see Definition 5.13). A non-invertible Rokhlin lemma is also developed by Rosenthal [317] in his work on topological models for measure-preserving systems and by Hoffman and Rudolph [155] in their extension of the Bernoulli theory to non-invertible systems.

⁽³⁵⁾(Page 63) This short proof comes from a paper of Wright [389], in which Kac's theorem is extended to measurable transformations.

⁽³⁶⁾(Page 64) The extension in Exercise 2.9.2 appears in the notes of Halmos [138, p. 71].

⁽³⁷⁾(Page 65) Exercise 2.9.3 is taken from a paper of Keane and Michel [188]; they also show that the supremum of $\mu(\bigcup_{i=1}^n T^{a_i}(A))$ over sets A for which

$$T^{a_1}(A), \dots, T^{a_n}(A)$$

are disjoint is a rational number, and show how this can be computed from the integers $a_1, \dots, a_n$.